STEVE SAMUEL

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EDUCATION

APJ Abdul Kalam Technological University **Kerala, India**

Bachelor of Technology in Computer Science and Engineering *2022 – 2026*

- **Relevant Coursework:** Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Machine Learning, Deep Learning, Software Engineering

EXPERIENCE

Data & Machine Learning Intern **June 2025**

NextGenPro Innovations and EduPark Pvt. Ltd. *Ernakulam, Kerala*

- Audited multi-class numerical and image datasets across feature distributions, identifying missing values, label anomalies, and extreme outliers using Python and NumPy.
- Engineered data cleaning and preprocessing routines, standardizing feature scales and formatting structured arrays to establish clean baseline data for downstream modeling pipelines.

AI Project Intern **Oct. 2024 – Dec. 2024**

Corizo (in collaboration with Wipro) *Remote*

- Evaluated baseline classification algorithms on structured clinical healthcare records across 13 physiological attributes, comparing model sensitivity to identify diagnostic risk patterns.
- Built an interactive desktop prototype application using Python and Tkinter, integrating feature scaling logic to accept patient metrics and output risk classifications in real time.

PROJECTS

FraudGuard AI | *Python, Scikit-learn, Random Forest, Streamlit* | GitHub | Live Demo **Oct. 2026**

- Built an imbalanced fraud detection pipeline on 284,807 transactions (0.172% fraud), tuning decision thresholds on validation data to maximize F1-score across class-weighted Random Forest and Logistic Regression models.
- Achieved 0.814 PR-AUC, 80.81% precision, 81.63% recall, and a 0.033% false-positive rate on a 56,962-transaction test partition at validation threshold 0.43; integrated MLflow tracking, automated Pytest suites, and a live Streamlit demo.

Automotive Image Classifier | *Python, TensorFlow, EfficientNetB0, Streamlit* | GitHub | Live Demo **June 2026 – Oct. 2026**

- Fine-tuned an EfficientNetB0 convolutional neural network for 196 fine-grained vehicle categories on the Stanford Cars dataset, standardizing 224x224 RGB inputs via transfer learning.
- Achieved 57.34% top-1 accuracy, 63.38% macro-precision, and 56.98% macro-F1 on a 1,629-image validation set; implemented a reproducible MLflow tracking workflow with an automated macro-F1 promotion gate and a deployed Streamlit demo.

IMDb Sentiment Analysis Platform | *Python, FastAPI, Docker, Streamlit* | GitHub | Live Demo **Oct. 2026**

- Developed an NLP classification pipeline across 50,000 IMDb movie reviews using custom negation-preserving text cleaning and sublinear TF-IDF (1, 2) n-grams, benchmarking Naive Bayes, Linear SVM, and Logistic Regression.
- Achieved 90.92% accuracy, 0.968 ROC-AUC, and 91.06% F1-score on 9,917 held-out test reviews; served single and batch CSV predictions via a FastAPI REST backend with Pydantic validation, containerized with Docker, and deployed on Streamlit Cloud.

TECHNICAL SKILLS

Programming & Query Languages: Python, Java, C, SQL, JavaScript

Data Science & Machine Learning: Scikit-learn, TensorFlow, Keras, EfficientNetB0, Random Forest, Logistic Regression, Transfer Learning, CNNs, NLP (TF-IDF), Imbalanced Learning

Backend, APIs & Development: FastAPI, Flask, RESTful APIs, Pydantic, Tkinter

Testing, DevOps & Tools: Git, GitHub Actions (CI), Pytest, Ruff, Docker, MLflow, Pandas, NumPy, Matplotlib, Seaborn

CERTIFICATIONS & WORKSHOPS

Robotics, AI & ML Workshop **June 2023**

Shristi Campus, Technopark *Trivandrum, India*

- Certified participant in technical workshop covering robotics fundamentals, artificial intelligence, and machine learning concepts.

AI & Machine Learning Program **Dec. 2024**

Corizo (in collaboration with Wipro) *Remote*

- Completed training program covering machine learning algorithms, data analysis, and predictive modeling project development.